1. Execution Plan Structure

1.1. Problem Definition and Objectives

- **Problem description:** The education system faces the challenge of keeping students present and regularly attending school. This project investigates how school infrastructure influences student attendance, using public data from the Ministry of Education (MEC) and INEP (School Census).
- **Project objectives:** The main objective is to investigate the correlation between school infrastructure and student attendance rates, identifying the most decisive factors that influence this aspect. In addition, the project aims to build a predictive model that can guide public policies on investment in school infrastructure to maximize student attendance, so that, when a renovation is planned, it is possible to know which infrastructure items are the best investments and will bring the greatest return in terms of student attendance.
- **Success metrics:** We will use the Accuracy metric, and our goal is to reach > 75%:
  - **Accuracy:** measures the proportion of correct predictions out of the total. Used in classification.
- **Stakeholders:** The main stakeholders include:
  - Ministry of Education (MEC): Interested in understanding the relationship between infrastructure and student attendance in order to adjust public policies.
  - State and municipal Education Departments: Responsible for implementing improvements to school infrastructure.
  - Educators and school administrators: Interested in practical recommendations for improving the school environment.
  - Students and their families: Direct beneficiaries of improvements to school infrastructure and of greater access to higher education.
  - Academic researchers: Involved in analyzing and disseminating the results.

1.2. Scope and Requirements

- **Data sources:**
  - **School Census 2024:** Up-to-date data on the infrastructure of schools in Brazil, available on the INEP open data portal. This dataset includes information on the physical infrastructure of schools, such as accessibility, available educational resources (such as internet, laboratories, etc.) and other aspects related to the school environment.
- **Known limitations:**
  - **Incomplete or outdated data:** Although we use the most recent School Census dataset (2024), there may be gaps in the data, such as schools without complete information about their infrastructure, or missing data.
  - **External factors:** Other factors, such as local education policies and differences in teaching practices, may influence ENEM results in addition to school infrastructure.

- **Scalability:** The project may face difficulties handling large volumes of data from different sources (ENEM, School Census, regional data) if the processing infrastructure is not sufficient.

- **Technical requirements:**
  - **Tools:**
    - Python (libraries such as pandas, numpy, matplotlib and seaborn) for data analysis, manipulation and visualization.
    - Jupyter Notebooks for developing exploratory analyses and creating interactive reports.
- **Required skills:**
  - **Data analysis:** Handling large volumes of data, identifying patterns and correlations.
  - **Statistics:** For analyzing the correlation between school infrastructure and ENEM performance.
  - **Machine Learning:** For training and using predictive models to understand the impact of different infrastructure variables.
  - **Data communication:** Ability to translate analysis results into clear, data-driven policy recommendations.
  - **Data management:** Use of ETL tools to integrate data from different sources and prepare it for analysis.
- **Deliverables:**
  - **Jupyter Notebook:** A notebook containing the whole research process, exploratory analysis, preprocessing, modeling and evaluation for training the model.
  - **Presentation:** In-class presentation of the final project results.

1.3. Project Stages

1. **Business Understanding**
   - Research in articles, MEC/INEP reports and academic literature
   - Clear definition of the problem and objectives
   - Establishment of success metrics (technical and business)
2. **Data Understanding**
   - Collection of the ENEM 2024 and School Census 2024 microdata
   - Study of the variable dictionary and metadata
   - Initial descriptive statistical analysis (absenteeism and performance by region)
   - Identification of inconsistencies and gaps in the data
3. **Data Preparation**
   - Cleaning and standardization of variables
   - Handling of missing values and encoding of categories
   - Creation of new features.
   - Splitting the data into training, validation and test sets
4. **Modeling**
   - Selection of candidate algorithms (classification and regression)

- Initial training (baseline)
- Performance comparison with technical metrics (AUC, recall, R^2)
- Hyperparameter optimization and refinement of the final model

5. **Evaluation**
   - Validation of the final model with business metrics
   - Comparison of the results with the research hypotheses
6. **Deployment**
   - Preparation of the final presentation
   - Delivery of the consolidated notebook

1.4. Schedule and Milestones

Schedule

- **Sprint 2 (10/14): Complete preprocessing**
  - Completion of data preprocessing, including cleaning, handling of missing values and preparation for modeling.
- **Sprint 3 (10/28): Initial modeling and feature engineering study**
  - Implementation and evaluation of initial models using the preprocessed data, testing different machine learning algorithms and starting the feature engineering study.
- **Sprint 4 (11/06): Optimization and refinement**
  - Evaluation of new features, organization of all project artifacts and saving the trained models.
- **Sprint 5 (11/25): Wrap-up and presentation preparation**
  - Consolidation of the final results, model adjustments, preparation of a final report and creation of a presentation.

Milestones

- **09/25: Planning and initial setup**
  - Definition of the project scope, objectives and schedule. Setup of the work environment and tools, and initial data collection.
- **10/02: Execution Plan delivery**
  - Delivery of the execution plan document, detailing the project stages and schedule, including objectives and required resources.
- **10/12: Approval of preprocessing and exploratory analysis**
  - Review and approval of the data cleaning process and the initial exploratory analysis performed.
- **10/21: Completion of initial modeling**
  - Completion of the first modeling tests and analysis of the preliminary results, including evaluation of model performance.
- **11/02: Review of optimization results**
  - Meeting to review the performance of the optimized models and make final adjustments before delivery.
- **11/25: Notebook consolidation and final fixes**

    - Merge the notebook into a single file and make the final adjustments needed to the model.
- **11/29: Script and slides creation**
    - Meeting to create the presentation script and slides.
- **12/02: Final presentation**
    - Present the work done throughout the semester.

1.5. Risks and Mitigations

Risk	Probability	Impact	Mitigation	Contingency plan
Insufficient data quality	Low	High	Perform an initial data assessment, plan effort for cleaning and feature engineering	If the data remains insufficient, look for alternative sources or complement it with external data
Model performance below expectations	Medium	High	Test multiple algorithms and invest in feature engineering	Tune hyperparameters and explore different modeling techniques, such as ensembles or additional tuning
Inconsistencies or gaps in the data	Low	Medium	Initial data validation, handling of missing values and standardization	If significant gaps are found, look for additional data or adjust the project scope
Lack of alignment between school infrastructure and impact variables	Medium	High	Refine the infrastructure-related variables and perform detailed qualitative analyses	Adjust the analysis iteratively, bringing in more qualitative data or field experts to redefine the variables
Legal restrictions and data privacy	Low	High	Ensure the data used complies with the LGPD (Brazilian data protection law), using public or anonymized data	If legal issues arise, stop using sensitive data and look for alternatives such as anonymized or aggregated data
Time constraints for analysis and delivery of results	Low	High	Set a realistic schedule with well-defined milestones and deliverables	Review the schedule and focus on priority tasks, adjusting the scope if necessary to ensure delivery of the main results
Memory and processing limitations in Google Colab	Medium	High	Optimize the algorithms and reduce memory usage with sampling or parallel processing techniques	If Colab cannot handle the data volume, split the dataset into smaller parts or run the project locally on our machines

Risk	Probability	Impact	Mitigation	Contingency plan
Connectivity failures and data loss in Google Colab	Medium	High	Periodically save data and models to Google Drive or another backup solution	If the session disconnects or the Colab environment loses the data, restore from the most recent backup and resume execution

1.6. Governance and Communication

- **Roles and responsibilities:** Each group member will have specific responsibilities in the project, as detailed below:
  Name: José Carlos Neto
  Responsibilities: Visualization, Research

  Name: Guilherme Cunha
  Responsibilities: Data Engineering, ML Modeling

  Name: Luidgi Colimerio
  Responsibilities: ML Modeling, Research

  Name: Lucio Oliveira
  Responsibilities: Documentation, Visualization

  Name: Rafael Estevao
  Responsibilities: Data Engineering, Documentation

- **Decision-making process:**
  - Decisions will be made after weekly meetings held in person or on Discord. During these meetings, members will discuss the technical and strategic issues of the project, taking everyone's contributions into account. The meeting leader (rotating) will be responsible for ensuring that discussions are concluded by consensus or, when necessary, by vote.
- **Communication frequency:**
  - Weekly meetings: Meetings will be held weekly, either in person or on Discord, to discuss project progress, align actions and make decisions.
  - Progress reports: Progress reports will be written weekly in the notebook provided for sprint tracking, making this artifact easier to control.
  - Informal updates: Quick updates can be shared more frequently in the WhatsApp group, as needed, to keep all project members aligned.
- **Communication channels:**
  - The WhatsApp group will be the main communication channel for quick messages, updates and discussion of specific points. Discord will be used for weekly alignment meetings and deeper discussions. Email will be used to send formal documents and more detailed information, or when it is necessary to share more complex materials (e.g., reports or deliverables).